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Lab Session 5

Python Strings — Lab Manual Content

Overview of the String Data Type

In Python, a **string** is a sequence of characters enclosed in quotes. Strings are one of the most commonly used data types and are **immutable** — meaning once created, their content cannot be changed in place. Strings can hold letters, digits, symbols, spaces, and even emojis. They are of type `str`.

Single-Line and Multi-Line Strings

Single-line strings are defined using single (`'`) or double (`"`) quotes on one line.

python

```
name = 'Yousif'  
greeting = "Hello, World!"
```

Multi-line strings span multiple lines and are defined using triple quotes (`' ''` or `"""`).

python

```
message = """ This is line one.  
This is line two.  
This is line three."""  
poem = '''Roses are red,  
Violets are blue,  
Python is awesome,  
And so are you.'''
```

Triple-quoted strings preserve all whitespace and newlines exactly as written.



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Common String Functions & Methods

Method / Function	Description	Example
<code>len(s)</code>	Returns the number of characters	<code>len("hello") → 5</code>
<code>s.lower()</code>	Converts all characters to lowercase	<code>"HELLO".lower() → "hello"</code>
<code>s.upper()</code>	Converts all characters to uppercase	<code>"hello".upper() → "HELLO"</code>
<code>s.capitalize()</code>	Capitalizes the first character only	<code>"python".capitalize() → "Python"</code>
<code>s.title()</code>	Capitalizes the first letter of each word	<code>"hello world".title() → "Hello World"</code>
<code>s.strip()</code>	Removes leading and trailing whitespace	<code>" hi ".strip() → "hi"</code>
<code>s.replace(old, new)</code>	Replaces occurrences of a substring	<code>"cat".replace("c", "b") → "bat"</code>
<code>s.split(sep)</code>	Splits the string into a list by separator	<code>"a,b,c".split(",") → ["a", "b", "c"]</code>
<code>s.find(sub)</code>	Returns the index of the first occurrence, or <code>-1</code>	<code>"hello".find("l") → 2</code>



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`s.count(sub)`

Counts how many times a
substring appears

`"banana".count("a") → 3`

`s.startswith(sub
)`

Returns **True** if the string
starts with the value

`"Python".startswith("Py") →
True`

`s.endswith(sub)`

Returns **True** if the string ends
with value

`"hello".endswith("lo") →
True`

Worked Examples (With Solutions)

Example 1 — Greeting with Length

Write a program that asks the user for their name, greets them, and tells them how many characters their name has.

python

```
name = input("Enter your name: ")
greeting = "Hello, " + name + "!"
print(greeting)
print("Your name has", len(name), "characters.")
```

Expected Output:

Enter your name: Yousif

Hello, Yousif!

Your name has 6 characters.



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Example 2 — Casing Transformations

Develop a script that prompts for a sentence and outputs the string in both fully capitalized and lowercase formats.

python

```
sentence = input("Enter a sentence: ")  
print("Uppercase:", sentence.upper())  
print("Lowercase:", sentence.lower())
```

Expected Output:

Enter a sentence: Hello World

Uppercase: HELLO WORLD

Lowercase: hello world

Example 3 — Substring Replacement

Construct a program that locates every space in a given sentence and replaces it with an underscore symbol.

python

```
sentence = input("Enter a sentence: ")  
new_sentence = sentence.replace(" ", "_")  
print("Modified:", new_sentence)
```

Expected Output:

Enter a sentence: Python is fun



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Modified: Python_is_fun

Example 4 — Character Frequency Analysis

Design a program to determine the total count of the letter 'a' within a user-provided word, ensuring the check is case-insensitive.

python

```
word = input("Enter a word: ")  
count = word.lower().count("a")  
print("The letter 'a' appears", count, "time(s).")
```

Expected Output:

Enter a word: Banana

The letter 'a' appears 3 times.

Example 5 — Generating Multi-Line Displays

Write code to render a descriptive user profile by utilizing the formatting power of multi-line strings.

python

```
name = "Yousif"  
age = 22  
city = "Karachi"  
  
profile = f"""  
--- User Profile ---
```



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```
Name : {name}
```

```
Age : {age}
```

```
City : {city}
```

```
_____ " " "
```

```
print(profile)
```

Expected Output:

```
--- User Profile ---
```

```
Name: Yousif
```

```
Age: 22
```

```
City: Karachi
```

```
-----
```

Practice Challenges (Exercise Only)

Problem 1 — Automatic Title Casing

Request the user's full name in any arbitrary case format. Your program should display the name correctly standardized using title case.

Expected Output:

```
Enter your full name: John Doe Smith
```

```
Formatted Name: John Doe Smith
```



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Problem 2 — Vocabulary Count Task

Prompt for a sentence from the user. Use string splitting to calculate and display the total number of words present.

Expected Output:

Enter a sentence: I love programming in Python

Total words: 5

Problem 3 — Palindrome Verification

Accept a word from the user and determine if it remains identical when reversed. Output the conclusion of the check.

Helpful Tip: Reversing can be achieved via slicing with `word[::-1]`

Expected Output:

Enter a word: madam

"madam" is a palindrome!

Enter a word: python

"python" is NOT a palindrome.

Problem 4 — Parsing Email Domains

Request an email address input. Successfully isolate and print the domain portion, representing all text following the @ character.

Helpful Tip: Consider using `.split("@")` to generate a list.



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Expected Output:

Enter your email: `yousifcreates@gmail.com`

Domain: `gmail.com`

Problem 5 — Comprehensive Text Analysis

Prompt for a sentence and return the following analytical metrics:

- The input string is converted to uppercase
- The overall character count, whitespace included
- The frequency of the character 'e' (ignoring case)
- Validation of whether the sentence initiates with the word "The."

Expected Output:

Enter a sentence: `The early bird catches the worm`

Uppercase : `THE EARLY BIRD CATCHES THE WORM`

Total characters: `30`

Count of 'e': `4`

Starts with 'The': `True`